Contents

COPYRIGHTED MATERIAL

Acknowledgments xv

Introduction xvii

Notation xxiii

Chapter 1 The Map and the Territory 1
1.1 The Securities 3
1.2 Modes of Exchange 5
1.3 Who Are the Market Participants? 6
1.3.1 The Sell Side 6
1.3.2 The Buy Side 9
1.4 Where Do Excess Returns Come From? 12
1.5 The Elements of Quantitative Investing 15

Chapter 2 Univariate Returns 20
2.1 Returns 21
2.1.1 Definitions 21
2.1.2 Excess Returns 23
2.1.3 Log Returns 23

2.1.4 Estimating Prices and Returns 24
2.1.5 Stylized Facts 26
2.2 Conditional Heteroskedastic Models 30
2.2.1 GARCH(1, 1) and Return Stylized Facts 32
2.2.2 GARCH as Random Recursive Equations 34
2.2.3 ★GARCH(1, 1) Estimation 36
2.2.4 Realized Volatility 37
2.3 State-Space Estimation of Variance 40
2.3.1 Muth's Original Model: EWMA 40
2.3.2 ★The Harvey–Shephard Model 44
2.4 ★Appendix 46
2.4.1 The Kalman Filter 46
2.4.2 Kalman Filter Examples 49
2.5 Exercises 51

Chapter 3 Interlude: What Is Performance? 53
3.1 Expected Return 54
3.2 Volatility 54
3.3 Sharpe Ratio 55
3.4 Capacity 58

Chapter 4 Linear Models of Returns 61
4.1 Factor Models 62
4.2 Interpretations of Factor Models 65
4.2.1 Graphical Model 66
4.2.2 Superposition of Effects 66
4.2.3 Single-Asset Product 67
4.3 Alpha Spanned and Alpha Orthogonal 68
4.4 Transformations 71
4.4.1 Rotations 71
4.4.2 Projections 73
4.4.3 Push-Outs 74
4.5 Applications 75
4.5.1 Performance Attribution 75
4.5.2 Risk Management: Forecast and Decomposition 76

4.5.3 Portfolio Management 80
4.5.4 Alpha Research 80
4.6 Factor Models Types 81
4.7 ⋆Appendix 82
4.7.1 Linear Regression 82
4.7.2 Linear Regression Decomposition 86
4.7.3 The Frisch–Waugh–Lovell Theorem 87
4.7.4 The Singular Value Decomposition 89
4.8 Exercises 92

Chapter 5 Evaluating Risk 94
5.1 Evaluating the Covariance Matrix 95
5.1.1 Robust Loss Functions for Volatility Estimation 95
5.1.2 Application to Multivariate Returns 97
5.2 Evaluating the Precision Matrix 100
5.2.1 Minimum-Variance Portfolios 100
5.2.2 Mahalanobis Distance 101
5.3 Ancillary Tests 102
5.3.1 Model Turnover 103
5.3.2 Testing Betas 103
5.3.3 Coefficient of Determination? 104
5.4 ⋆Appendix 107
5.4.1 Proof for Minimum-Variance Portfolios 107

Chapter 6 Fundamental Factor Models 110
6.1 The Inputs and the Process 111
6.1.1 The Inputs 111
6.1.2 The Process 114
6.2 Cross-Sectional Regression 115
6.2.1 Rank-Deficient Loadings Matrices 118
6.3 Estimating the Factor Covariance Matrix 120
6.3.1 Factor Covariance Matrix Shrinkage 121
6.3.2 Dynamic Conditional Correlation 122

6.3.3 Short-Term Volatility Updating 122
6.3.4 Correcting for Autocorrelation in Factor Returns 124
6.4 Estimating the Idiosyncratic Covariance Matrix 125
6.4.1 Exponential Weighting 125
6.4.2 Visual Inspection 125
6.4.3 Short-Term Idio Update 126
6.4.4 Off-Diagonal Clustering 127
6.4.5 Idiosyncratic Covariance Matrix Shrinkage 131
6.5 Winsorization of Returns 131
6.6 ⋆Advanced Model Topics 133
6.6.1 Linking Models 133
6.6.2 Currency Rebasing 139
6.7 A Tour of Factors 141

Chapter 7 Statistical Factor Models 147
7.1 Statistical Models: The Basics 149
7.1.1 Best Low-Rank Approximation and PCA 149
7.1.2 Maximum Likelihood Estimation and PCA 152
7.1.3 Cross-Sectional and Time-Series Regressions via SVD 155
7.2 Beyond the Basics 155
7.2.1 The Spiked Covariance Model 156
7.2.2 Spectral Limit Behavior of the Spiked Covariance Model 158
7.2.3 Optimal Shrinkage of Eigenvalues 160
7.2.4 Eigenvalues: Experiments versus Theory 162
7.2.5 Choosing the Number of Factors 162
7.3 Real-Life Stylized Behavior of PCA 165
7.3.1 Concentration of Eigenvalues 166
7.3.2 Controlling the Turnover of Eigenvectors 168

7.4 Interpreting Principal Components 173
7.4.1 The Clustering View 173
7.4.2 The Regression View 174
7.5 Statistical Model Estimation in Practice 176
7.5.1 Weighted and Two-Stage PCA 176
7.5.2 Implementing Statistical Models in Production 179
7.6 ⋆Appendix 181
7.6.1 Exercises and Extensions to PCA 181
7.6.2 Asymptotic Properties of PCA 185

Chapter 8 Evaluating Excess Returns 188
8.1 Backtesting Best Practices 190
8.1.1 Data Sourcing 190
8.1.2 Research Process 191
8.2 The Backtesting Protocol 195
8.2.1 Cross-Validation and Walk-Forward 195
8.3 The Rademacher Anti-Serum (RAS) 200
8.3.1 Setup 200
8.3.2 Main Result and Interpretation 203
8.4 Some Empirical Results 208
8.4.1 Simulations 208
8.4.2 Historical Anomalies 210
8.5 ⋆Appendix 214
8.5.1 Proofs for RAS 214

Chapter 9 Portfolio Management: The Basics 220
9.1 Why Mean-Variance Optimization? 221
9.2 Mean-Variance Optimal Portfolios 223
9.3 Trading in Factor Space 229
9.3.1 Factor-Mimicking Portfolios 229
9.3.2 Adding, Estimating, and Trading a New Factor 232
9.3.3 Factor Portfolios from Sorts? 235
9.4 Trading in Idio Space 236
9.5 Drivers of Information Ratio: Information Coefficient and Diversification 237
9.6 Aggregation: Signals versus Portfolios 240

9.7 ⋆Appendix 244
9.7.1 Some Useful Results from Linear Algebra 244
9.7.2 Some Portfolio Optimization Problems 245
9.7.3 Optimality of FMPs 245
9.7.4 Single-Factor Covariance Matrix Updating 247

Chapter 10 Beyond Simple Mean-Variance 250
10.1 Shortcomings of Naïve MVO 251
10.2 Constraints and Modified Objectives 254
10.2.1 Types of Constraints 257
10.2.2 Do Constraints Improve or Worsen Performance? 261
10.2.3 Constraints as Penalties 261
10.3 How Does Estimation Error Affect the Sharpe Ratio? 267
10.3.1 The Impact of Alpha Error 268
10.3.2 The Impact of Risk Error 269
10.4 ⋆Appendix 270
10.4.1 Theorems on Sharpe Efficiency Loss 270

Chapter 11 Market-Impact-Aware Portfolio Management 276
11.1 Market Impact 277
11.1.1 Temporary Market Impact 278
11.2 Finite-Horizon Optimization 284
11.3 Infinite-Horizon Optimization 286
11.3.1 Comparison to Single-Period Optimization 289
11.3.2 The No-Market-Impact Limit 290
11.3.3 Optimal Liquidation 290
11.3.4 Deterministic Alpha 291
11.3.5 AR(1) Signal 291
11.4 ⋆Appendix 293
11.4.1 Proof of the Infinite-Horizon Quadratic Problem 293

Chapter 12 Hedging 297
12.1 Toy Story 298
12.2 Factor Hedging 301
12.2.1 The General Case 301
12.3 Hedging Tradeable Factors with Time-Series Betas 304
12.4 Factor-Mimicking Portfolios of Time Series 307
12.5 ⋆Appendix 309

Chapter 13 Dynamic Risk Allocation 312
13.1 The Kelly Criterion 314
13.2 Mathematical Properties 321
13.3 The Fractional Kelly Strategy 323
13.4 Fractional Kelly and Drawdown Control 327

Chapter 14 Ex-Post Performance Attribution 333
14.1 Performance Attribution: The Basics 335
14.2 Performance Attribution with Errors 336
14.2.1 Two Paradoxes 336
14.2.2 Estimating Attribution Errors 337
14.2.3 Paradox Resolution 339
14.3 Maximal Performance Attribution 340
14.4 Selection versus Sizing Attribution 347
14.4.1 Connection to the Fundamental Law of Active Management 351
14.4.2 Long–Short Performance Attribution 351
14.5 Appendix⋆ 352
14.5.1 Proof of the Selection versus Sizing Decomposition 352

Chapter 15 A Coda about Leitmotifs 357

References 359
Index 373